

Reciprocal relationships, reverse causality, and temporal ordering: Testing theories with cross-lagged panel models

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Abstract from panel; we may want to decide on specific illustrative case after we've laid out framework of paper, as we want our case to exhibit every potential issue.

Reciprocal causal relationships are a common feature of criminological theories. For example, police forces tend to use force more often in areas where crime concentrates, while at the same time legal cynicism theory suggests that cumulative exposures to police use-of-force can foster criminal activity. When multiple observations over time are available, cross-lagged panel models are commonly used to estimate these reciprocal effects. This is often done without careful attention to the assumptions that must be satisfied to produce valid estimates, such as correctly specified temporal lags, sufficient inter-temporal variation, and proper accounting for unobserved heterogeneity. Failure to satisfy these assumptions can produce severe issues including spurious associations and parameter estimates that are biased or even reversed in direction. In addition, reciprocal relationships violate causal assumptions based on graphical tools; criminological theories that suggest reciprocal causal relationships usually have an underlying macro-micro mechanism often not accounted for in empirical models. We provide guidance on how to align theory, model specification, and choice of estimator and illustrate this using an empirical example. We use data from Chicago at the census tract level and model the potential reciprocal relationship between police use-of-force and violent crime between 2004 and 2016. We finalise highlighting the importance of criminological theory and careful attention to empirical implications of theoretical premises when investigating reciprocal relationships

Introduction

- Initial hook
 - Tell you how to use a CLPM
 - When do you use a CLPM? [DO this later in detail]

- Use it when you expect reciprocal effects
- Also within and between, but NOT our focus; refer to XXX
- Only dynamic models, not static models – though this is important for static models as well
- Explanation of how what we’re doing is different from recent reviews of CLPM literature, e.g., Zyphur et al. (2020).
 - * Zyphur et al. elaborate a complete generalized cross-lagged panel model that accounts for a wider range of specifications. Here we focus on key problems that appear in applied literature.
- Theory and reciprocity
 - When do we expect reciprocal effects? In context of theory testing
- The role of time
 - What are reciprocal effects *really*?
 - * There are no simultaneous effects
 - * Usually a time problem, either:
 - Ambiguity from theory (i.e., reciprocity is substantive)
 - Competing theory when they are specific (i.e., substantive, empirical adjudication problem)
 - Purely empirical problem due to repeated observations (i.e., reciprocity is a nuisance)
 - Separate theoretical and methodological concerns
 - Strong theories should be specific about timing in causal processes; e.g., alcohol to disinhibition; well-defined, observed biological process (a pint today may give you a headache tomorrow but won’t reduce self-control; zero contemporaneous headache).
- Introduce the CLPM
 - What must a model do to capture these?
 - Use 3 wave model as example

Two motivating examples

To illustrate our points, we consider two canonical longitudinal relationships in criminology. The first is the link between employment (occupational attainment) and offending—a focus of classic life-course research (e.g., [Sampson and Laub 1992, 1993](#)). The second is the link between residents’ perceptions of neighborhood disorder and their fear of crime—a central topic for neighborhood scholars ([Sampson and Raudenbush 2004](#); [Brunton-Smith 2011](#)). In the employment–crime example, theory and policy concern center on the effect of employment on future offending, whereas in the fear–disorder example the direction of causation is itself debated.

Employment and crime

Life-course studies routinely find that obtaining stable employment tends to reduce criminal behavior (Ramakers et al. 2017; Lageson and Uggen 2012). The logic is straightforward. In the short run, being employed fills one’s time with routine activities and conventional social bonds, which constrains opportunities and incentives for offending (Cohen and Felson 2010; Hirschi 2017). In the long run, it is not employment per se but stable, committed work that enhances social control and thereby reduces crime (Sampson and Laub 1993). In other words, the accumulation of conventional ties and responsibilities that are often associated with stable employment provides identity and that deter future offending (Ramakers et al. 2017). For example, longitudinal reviews show that employed individuals have significantly lower recidivism rates than the unemployed (Uggen and Wakefield 2008). Indeed, some experimental job-program evaluations have found that random assignment to job training can yield lower subsequent crime (Redcross et al. 2011).

Crucially, the reverse causal path is also possible. After all, criminal arrests or behavior can lead to job loss. Therefore, when estimating the effects on employment on crime, one needs to account for reverse causality. The crime $\rightarrow$ employment relationship is not of immediate substantive interest to most criminologists, so in this example we regard that as an endogeneity problem. Our substantive interest is in the effect of employment on later crime, treating any effects of crime involvement on occupational attainment as a nuisance and a source of bias which needs to be controlled for.

Fear of crime and perceived disorder

By contrast, the relationship between perceived disorder and fear of crime is typically viewed as reciprocal. Some theories posit that visible disorder in the environment raises fear of crime—disorderly cues signal that an area is in decline and unable to control deviance (Kelling, Wilson, et al. 1982). Others emphasize the opposite interpretation: individuals with high levels of fear of crime are sensitized to local disorder, causing them to perceive and interpret ambiguous cues as physical and social disorder (Sampson and Raudenbush 2004; Jackson 2004). In practice, both processes may occur (Brunton-Smith 2011).

Fear and disorder perceptions are also rapidly-occurring updating processes. Given the theoretical ambiguity and the possibility for reciprocal relationships over time, this example does not have clearly defined dependent and independent variables. Both fear of crime and perceived disorder are an outcome as well as a predictor of interest. Our substantive interest is therefore the reciprocity itself. We can use panel models either to adjudicate between competing theories or to discover actual reciprocal effects over time.

Both examples are contrasted by how they are framed theoretically. In the employment–crime case, the cross-lagged panel model is used mainly to control for the reverse effect (crime $\rightarrow$ job) and isolate the job $\rightarrow$ crime effect of interest. In the fear–disorder case, by contrast,

both lagged paths are of substantive interest, reflecting genuine theoretical ambiguity. Which direction we emphasize is dictated by the research question. In the first example we treat the reverse path as a nuisance (a classic reverse-causality scenario), whereas in the second we treat it as part of a mutually reciprocal process to be tested (i.e., theoretical reciprocity).

Theoretical and statistical models

Good empirical research starts with a strong theoretical foundation. It is from a clearly defined theoretical framework that observable implications can be deduced, and it is those observable implications which can then be contrasted against empirical data (King, Keohane, and Verba 1994; Mayo 2018). Theory-testing empirical research is therefore conditional on a thorough and transparent theoretical model (Nagin and Sampson 2019). Given that most criminological theories presume causal processes, the first step should normally involve deriving a non-recursive model that accurately represents the theorized relationships—e.g., the theoretical model could be represented by a directed acyclic graph (DAG). DAGs can formally represent a nonparametric structural equation model, avoiding functional-form and distributional assumptions (Pearl 2009), and are increasingly used to portray premised causal relationships in criminological theory (Lanfear, Matsueda, and Beach 2020; Nivette and Vegt 2025; Oliveira 2024). The DAG should be all-encompassing and include all relevant theorized pathways, including unobserved mechanisms.

Once the theoretical model is specified, it should guide the selection of an appropriate estimator. It is important to distinguish the theoretical model from the statistical model: while the latter should approximate the former as much as possible, the statistical model will unavoidably have to handle imprecision and uncertainty that are bound to exist when dealing with empirical data, such as distributional and parametric assumptions and unexplained covariances due to theoretical ambiguities. When dealing with situations involving potential reciprocal relationships and/or reverse causality, we propose leveraging DAGs for theoretical models and drawing on the structural equation modeling (SEM) framework for estimation (Bollen and Pearl 2013). A clear theoretical model is a necessary prerequisite for specifying an estimator. While both diagrams will not be exactly the same, we suggest that the premised DAG inform some elements of the SEM. Specifically, uni-directional arrows in the structural model should be non-recursive, as per the theoretical model.¹ For example, a key consideration in this process is the timing of effects: theorized effects can be contemporaneous (i.e., occur quickly) or lagged (i.e., unfold more gradually over time). Yet, while the theoretical model displays a non-parametric representation of theorized causal relationships, the SEM

¹SEMs include a measurement component, which specifies measurement models linking manifest indicators and latent variables, and a structural component, which specifies the relationships between latent variables. For now, we are exclusively referring to the structural component of an SEM. Those include bi-directional arrows, which depict the estimation of a covariance between (the error terms of) two variables, and one-directional arrows, which indicate the estimation of directional association based on regression modeling.

can include bi-directional arrows representing empirically observed covariances between variables not explained by the premised DAG—making the statistical model a recursive model. Simultaneous equations should only be used only as a last resort, given their often tenuous theoretical grounding and the risk of overfitting.

When is a CLPM needed?

A cross-lagged panel model is only needed when reciprocal effects over time are presumed in the theoretical model. When the researcher can theoretically assume no effects of the dependent variable on the independent variable over time, other statistical models will be more appropriate than a cross-lagged panel model (see [Imai and Kim 2019](#)). For example, research focused on the effects of a single-event exposure—a common topic in life-course criminology and the role of turning points, such as the long-term effects of school drop-out or marriage on offending ([Sampson and Laub 2016](#); [Nguyen and Loughran 2018](#))—would not benefit from cross-lagged panel models.

Instead, reciprocal effects over time are most often a threat when the exposure is a continuous process that happens repeatedly. There are two situations in which reciprocal effects could emerge from theoretical models, thus highlighting the need of cross-lagged panel models. We name them *reverse causality* and *theoretical reciprocity*.

Reverse causality refers to situations in which researchers are primarily interested in assessing the effects of an exposure on an outcome but are concerned about the possibility that the outcome also influences future exposures. From our empirical example, assessing the effects of occupational attainment on crime involvement is of particular theoretical and policy relevance ([Sampson and Laub 1993](#)); in order to assess that relationship, however, one needs to take into account that offending behavior may also have an effect on future employment, even though examining prospects of job loss following a criminal arrest or behavior is not of substantive interest. This is a classic endogeneity problem, in which not controlling for reverse causality introduces bias. This has been a long concern of econometricians—e.g., the Arellano-Bond estimator, which is essentially a cross-lagged panel model, was developed specifically to control for reverse causality ([Arellano and Bond 1991](#); [Moral-Benito, Allison, and Williams 2019](#)). When dealing with reverse causality, researchers have a predefined exposure and an outcome of interest and simply aiming to control away the effects of the outcome on the exposure over time.

What we call theoretical reciprocity, on the other hand, refers to situations in which the directionality of the effects is ambiguous. This could be due to, for instance, competing theories suggesting opposing pathways between two variables ([Sturgis et al. 2004](#)). From our empirical example, some researchers argue that perceived disorder leads to increases in fear of crime ([Kelling, Wilson, et al. 1982](#)) whereas others approach this relationship from the opposite direction, arguing that fear of crime leads local residents to view disorderly signs in the neighborhood as problematic ([Sampson and Raudenbush 2004](#); [Jackson 2004](#)). This theoretical

ambiguity implies that reciprocity is an empirical problem—a question that can be adjudicated by theory testers (Brunton-Smith 2011). When dealing with theoretical reciprocity, researchers often do not have a predefined exposure and an outcome of interest and instead aim to assess both directions.

In both cases, the theoretical model will highlight the threat of reciprocal relationships. A DAG will not distinguish between reciprocal effects due to theoretical reciprocity or due to reverse causality; this distinction is purely conceptual and should be dictated by the research question. Yet, we sustain that this is an important distinction. While in some cases the empirical solution will be similar, in other situations dealing with theoretical reciprocity or reverse causality may lead to different statistical estimators—we expand on this point below.

CHUCK

Imaginary data

In both cases we have three waves of panel data measured in yearly waves.

- For example 1, we know the updating happens much faster than over course of a year; even if we had daily data, it would still be too fast. COuld do it with lab equipment, but lose real world construct and external validity.
- For example 2, is not as fast, but over course of year, still plenty of time for feedback; if we decrease gaps between waves, we'd begin to rule out reverse causality. Theory can imply how narrow a time period lets you rule out.
- Reality and theory move at one recursive (really continuous) pace, but we get snapshots of data

FIGURE: Time DAG

- Continuous time means you can imagine an infinite number of these “missing waves”; don't be too worried about that, but do read this stuff

SEM diagrams

- SEM path diagrams for estimation
 - The structural part of the SEM should also be a DAG, except in rare circumstances, e.g., pure simultaneous equations; these require very strong identifying assumptions.
 - Work out the diagrams; see if they are the same? Probably are!
- Data observed, and thus you must estimate, from discrete often arbitrarily spaced waves; this produces reverse causality

FIGURE: Time estimator; the classic CLPM

- Explain the classic CLPM as an estimator
 - Strict exogeneity

Thiago: Bollen & Pearl 2009 provide a nice discussion about the notational relationship between SEM path diagrams and DAGs. We should also look at Imai & Kim's three papers on causal inference with panel data. They use DAGs and they're so well written. I don't think they discuss reciprocal effects at all (could be wrong), but we could rely on them from a notational point of view in using DAGs to depict temporal processes.

How it can go wrong!

- One para introducing the three key issues
- DIVVY UP AFTER

Issue 1: Temporal Order

- Getting lag specification wrong is a big issue
- This assumes true or approximately true lag specification
- Unusual form of bias as documented by Vaisey & Miles that can flip signs:
 - If true model is $y = \beta x_t + \alpha_i + e_{it}$ and you fit $y = \beta^* x_{t-t} + \alpha_i + e_{it}$, $E(\beta^*) = -0.5\beta$, i.e., as Allison (2022) says “a positive contemporaneous effect of x gets transformed into an artifactually negative lagged effect”.
- Probably the most underappreciated issue as it may produce misleading results that reject or overturn theories
 - Papers that find negative effects when theory suggests positive ones may be the result of this bias.
- Key takeaway: Don't just be worried about your research but also be skeptical about interpreting other research with unexpected sign flips!

Solutions

- Theoretical ideal: Get it right
- Vaisey & Miles: Do both lagged and contemporaneous
 - If no contemporaneous effect, only lagged, you're probably fine. If contemporaneous shows up, you can't definitively determine direction of contemporaneous effect.

- Allison: Partial correlation model with lagged effect and contemporaneous residual covariance
 - This is least biased approach if parameter of interest is βx_{t-1} , but cannot estimate βx_t
 - This still does not adjudicate between direction of contemporaneous effects; if effect is entirely contemporaneous, all relationship will be in residual covariance.
- Also good place to bring in predetermined regressors which are part of Allison approach and Arellano-Bond

Issue 2: Unobserved heterogeneity

- Probably the most commonly encountered issue; but also pretty much solved!
- Most common concern is stable traits
 - While really same problem, we treat time-varying and time-stable confounders as different problems in longitudinal models
 - Autoregressive term alone does not solve problem
 - Can't just toss in fixed effects due to Nickell bias from endogenous lag (shrinks in proportion to T); Arellano-Bond (and thus Allison) is removing Nickell bias
- Including time-stable covariates in FE models
- Might also raise treatment heterogeneity here as different but related issue not as easily addressed in CLPMs as in TWFE
 - Developmental theories often specify age-specific (life stage) treatment heterogeneity
 - Freeing the parameter by wave allows for time heterogeneity but also effects to differ because wave spacing is different
 - * One reason for treatment heterogeneity among individuals is differing measurement times within waves
 - Exception is time-specific treatment heterogeneity which is easily permitted by allowing wave-specific parameters as in Allison's model
 - A potentially useful application for wave-specific parameters is when you have unevenly spaced data; I think latent growth models would get at this too but I'm not very familiar with them.
- Predetermined regressors (weakly exogenous, sequentially exogenous)

Solutions

While the traditional cross-lagged panel model cannot properly account for unobserved stable heterogeneity, some relatively recent extensions of the CLPM yield robust estimators that draw only on within-unit variation and effectively control for all unobserved time-stable traits (Leszczensky and Wolbring 2019). Specifically, Allison, Williams, and Moral-Benito (2017)’s maximum likelihood structural equation fixed-effects dynamic panel method (ML-SEM) and Hamaker, Kuiper, and Grasman (2015)’s random intercept cross-lagged panel model (RI-CLPM) model reciprocity and account for unobserved, time-stable features that impact both Y_t and X_t .

Fixed-effects dynamic panel model with maximum likelihood

In ML-SEM (Allison, Williams, and Moral-Benito 2017), a standard dynamic panel model (with a lagged dependent variable) is recast as a structural equation model that explicitly partials out all unobserved stable heterogeneity. In practice, one introduces a latent “fixed-effect” factor (often denoted α) that loads with coefficient 1 on every observed Y_t for individual i . This latent random intercept factor is the same as the one often included in latent growth curve models, representing between-unit differences. By fixing the factor loadings of the random intercept to 1, the model effectively “demeans” each unit’s scores and thus conditions out all time-constant heterogeneity.

Crucially, this latent random intercept factor is allowed to freely covary with all the time-varying exogenous predictors (including any predetermined regressors and the initial outcome Y_{i1}). This model is exactly a correlated-random-effects model (Mundlak 1978): by letting α_i correlate without restriction with the X ’s, the SEM mimics a fixed-effects specification (the contrast being that a classical random-effects model would force α_i to be independent of X). As Allison, Williams, and Moral-Benito (2017) note, allowing α_i to covary with X ’s enforces a fixed-effects model that accounts for all unobserved, stable confounders. This means that the ML-SEM assumes a key outcome (e.g., Y) and a key predictor (e.g., X) of interest, and the whole is set in such a way so as to find effects of the predictor on the outcome (e.g., effects of X_{t-1} and/or X_t on Y_t) while accounting for potential effects of Y_{t-1} on X_t . If the effects of Y_{t-1} or Y_t on X_t are also of substantive interest, one needs to fit a separate ML-SEM with that specification.

In implementation, the entire system of autoregressive and cross-lagged equations is cast as a single SEM. For example, with four waves one writes each Y_{it} on its lagged value $Y_{i,t-1}$ and on lagged $X_{i,t-1}$, as well as on a latent factor α_i loading on each Y_{it} . Figure 1 illustrates this: α represents the latent random intercept factor as is reflected by all observations Y_t (except for the initial observation $Y_{t=1}$) with factor loadings constrained to 1, and all X_t are allowed to correlate with α (and with each other and with $Y_{t=1}$). In this setup, the first observation $Y_{t=1}$ is treated like any exogenous covariate. Importantly, observed time-varying regressors (X_t) are specified as predetermined—that is, they are allowed to correlate

with past residuals (i.e., with $\epsilon_{i,t-1}$), reflecting the idea that predictors may be influenced by prior outcomes. The model is equivalent to the Arellano-Bond method estimated with the generalized method of moments (Arellano and Bond 1991), but it requires no stationarity assumption on initial conditions (given that initial observations Y_1 and X_1 are simply treated as additional exogenous predictors). The ML-SEM also expands upon the Arellano-Bond method as it can handle missing data with full information maximum likelihood, it can easily extend to multiple indicators, and it can incorporate time-constant covariates and different timing effects. The ML-SEM model is a standard SEM with autoregressive and cross-lagged parameter using only within-unit variation and a latent random intercept α factor soaking up all between-unit variation. Figure 1 depicts the ML-SEM.

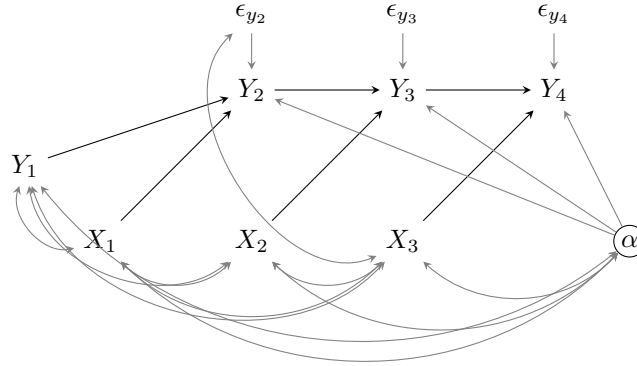


Figure 1: Maximum likelihood structural equation fixed-effects dynamic panel model (Allison et al., 2017)

Random intercept cross-lagged panel model

The other alternative to the traditional cross-lagged panel model that actually controls for unobserved stable heterogeneity is Hamaker, Kuiper, and Grasman (2015)’s Random Intercept Cross-Lagged Panel Model (RI-CLPM). To ensure that the model only takes into account change over time, the RI-CLPM explicitly models within-unit and between-unit variances as separate attributes. Similar to Allison, Williams, and Moral-Benito (2017)’s ML-SEM, the first step involves estimating a latent variable that captures all stable variation between units—again drawing on the latent growth curve model specification, this latent variable is defined by all observations Y_t over time with factor loadings constrained to 1, essentially creating a random intercept for Y that varies between units.

While Allison, Williams, and Moral-Benito (2017)’s approach involves allowing this latent variable to covary with predetermined regressors—which implies treating it as a fixed effect that captures all unobserved stable heterogeneity—the RI-CLPM instead repeats the process and creates another latent variable defined by all observations X_t over time, thus creating a random intercept for X that varies between units. Treating such unit-specific random intercepts as random variables (as opposed to fixed parameters) usually implies that they do not capture all

unobserved stable heterogeneity. However, in the RI-CLPM, the two latent variables reflecting random intercepts for Y and X (represented by α_Y and α_X in Figure Figure 2) are allowed to covary, which ensures that they capture all unobserved stable heterogeneity across both variables. Figure Figure 2 depicts the the RI-CLPM.

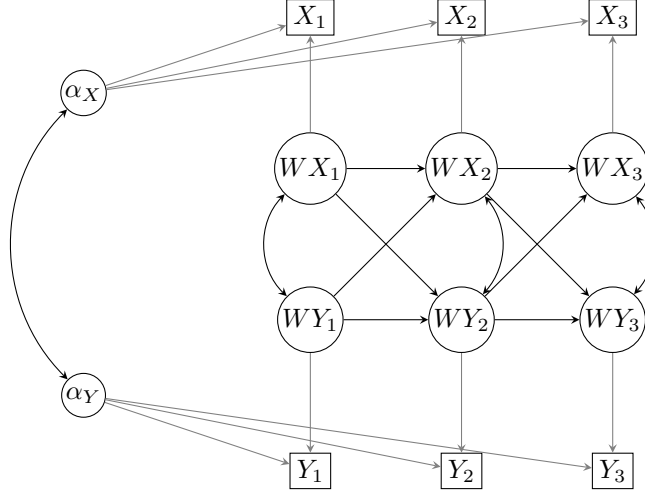


Figure 2: Random Intercept Cross-Lagged Panel Model (Hamaker et al., 2015)

To further ensure that only within-unit variation is taken into account, the RI-CLPM relies on a set of other latent variables. For each variable Y_t and X_t , a new latent variable is estimated with a factor loading constrained to 1. These latent variables are represented in the diagram in Figure Figure 2 as WY_t and WX_t . This implies that every single observed variable Y_t and X_t is defined by two latent variables: one capturing all differences between units (i.e., α_Y and α_X), and one capturing everything else, i.e., within-unit differences (i.e., WY_t and WX_t). Autoregressive and cross-lagged parameters are then included linking WY_t and WX_t —resembling the traditional cross-lagged panel model, but exclusively focused on within-unit variation over time. By allowing both random intercepts to covary, the RI-CLPM ensures that all unobserved stable heterogeneity is accounted for; and by estimating autoregressive and cross-lagged effects on latent variables capturing only within-unit variation, the model ensures that cross-lagged estimates reflect change over time. The RI-CLPM can easily be extended to other common features in SEMs, such as multiple indicators, covariate adjustment, and multiple-group analysis (Mulder and Hamaker 2021).

Issue 3: Inter-temporal variation

- Maybe show some illustrative math here to show that $Var(Y_2|Y_1) \rightarrow 0$ as $\rho(Y_1, Y_2) \rightarrow 1$ approaches 1; i.e., you rapidly run out of anything to explain.

- This is essentially a multicollinearity problem: Unbiased and consistent but high variance in estimates makes any particular estimate suspect.
 - From time series lit: Stronger your autoregressive term is, the less there is to explain; rapidly approaches a time-stable
- Rarely appreciated outside time series literature but common when waves are narrowly spaced
- With random measurement error, the error becomes proportionally larger as time periods get narrower
 - May mean you can address in part with measurement models
 - Note the endogenous variables are regressors, so measurement error biases parameter estimates as well
- Psychological constructs can be very stable over time! Life course trajectories are real!
 - People even get more consistent over time

Solutions

- This is first and foremost a data problem, so you want to attack it during design of data collection if possible
 - Sample to maximize change
 - Look for shocks
- Can put time stable covariates using hybrid approach, but be aware, you're now looking at between-individual effects
- If you can't collect new data, look at different aggregations or wave skips
 - Small change at a large unit may mask larger changes in small units
 - Should prob only consider skipping waves if it makes sense theoretically and/or you're facing serious empirical underidentification problems.
- If increasing error is what you're worried about, can potentially use measurement models
 - Should be doing this anyway because your outcomes are regressors, so measurement error attenuates estimates
- Give up and go get a pint

General Advice

- Separate theory problems from estimation problems
 - Theory comes first

- Then a clear model
- Then estimator
- Default to robust estimators
 - Allison’s FE model should probab be the default as it is most robust
- Also need to think about all these issues not specific to CLPM or reciprocal effects
 - Sequential ignorability
 - Dynamic effects
 - * Other related strategies, e.g., for single outcomes with continuous time-varying confounded exposures

Conclusion

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